

CFB VALCARTIER, QC, Canada

WMO: 717160

Lat: 46.9004N

Lon: 71.5032W

Elev: 550

StdP: 14.41

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.0	-16.9	-31.5	1.0	-21.8	-26.2	1.3	-16.3	20.3	22.4	18.2	18.5	1.7	230	0.583

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.2	83.2	68.6	80.3	67.1	77.6	65.5	71.8	79.0	69.8	76.3	68.0	74.4	9.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.5	110.5	75.3	67.6	103.4	73.5	65.6	96.4	71.8	36.0	78.9	34.2	76.4	32.7	74.5	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.2	15.9	13.9	DB	-32.9	88.2	6.6	2.4	-37.6	90.0	-41.5	91.4	-45.2	92.8	-50.0	94.6	(4)
(5)			WB	-32.8	74.8	6.5	2.5	-37.5	76.7	-41.4	78.1	-45.0	79.6	-49.8	81.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	38.8	8.5	11.2	22.0	36.6	51.1	60.3	65.4	63.3	55.0	43.2	30.3	16.6	(6)
(7)		DBStd	21.90	13.97	12.80	11.68	7.69	7.31	6.52	4.97	5.38	7.05	7.19	9.19	12.00	(7)
(8)		HDD50	5614	1288	1086	869	405	75	3	0	0	30	233	591	1034	(8)
(9)		HDD65	9727	1753	1506	1334	852	435	169	56	97	309	677	1040	1499	(9)
(10)		CDD50	1513	0	0	0	3	108	313	476	412	179	21	1	0	(10)
(11)		CDD65	151	0	0	0	0	3	29	67	44	8	0	0	0	(11)
(12)		CDH74	1524	0	0	0	2	101	374	586	373	88	0	0	0	(12)
(13)		CDH80	277	0	0	0	0	18	81	109	55	14	0	0	0	(13)
(14)	Wind	WSAvg	5.3	5.3	5.6	6.0	6.2	5.8	5.3	4.8	4.5	4.3	5.1	5.2	5.0	(14)
(15)	Precipitation	PrecAvg	48.1	3.4	2.8	2.9	3.3	3.7	4.4	5.6	4.6	4.8	4.6	3.8	4.1	(15)
(16)		PrecMax	55.4	5.6	4.6	5.2	7.6	6.1	8.2	9.4	6.9	9.1	9.2	6.2	8.9	(16)
(17)		PrecMin	38.9	1.5	1.1	0.8	0.7	2.0	2.0	2.8	1.5	2.4	1.6	1.4	1.8	(17)
(18)		PrecStd	4.8	1.2	0.9	1.0	1.7	1.1	1.7	1.7	1.2	1.7	1.7	1.1	1.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.0	41.0	50.9	67.4	82.5	86.3	87.3	84.5	82.0	69.9	56.7	41.8	(19)
(20)			MCWB	38.6	36.7	41.6	51.3	62.8	70.7	72.5	69.9	69.2	60.0	53.4	40.7	(20)
(21)		2%	DB	36.2	37.1	44.3	59.9	76.6	82.1	83.0	81.3	76.2	63.6	50.5	37.9	(21)
(22)			MCWB	34.7	33.3	37.2	47.0	59.5	67.3	69.3	68.3	65.7	56.2	47.1	36.2	(22)
(23)		5%	DB	33.1	34.1	40.5	54.3	71.5	78.5	79.9	78.2	72.0	58.9	46.2	35.3	(23)
(24)			MCWB	31.5	31.9	35.2	44.0	57.1	65.1	67.6	66.4	63.4	53.1	42.8	33.7	(24)
(25)		10%	DB	29.0	31.5	37.4	49.8	66.5	74.4	77.2	75.2	68.1	55.1	42.7	33.1	(25)
(26)			MCWB	27.2	29.3	33.6	41.7	54.6	62.5	66.1	64.7	61.7	50.5	40.3	31.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.1	38.5	43.9	56.2	67.4	73.9	75.0	72.8	71.1	62.2	54.7	40.9	(27)
(28)			MADB	39.8	39.9	47.4	65.4	76.8	82.7	83.1	81.0	77.5	67.3	56.1	41.3	(28)
(29)		2%	WB	34.8	34.1	38.9	48.8	62.9	70.0	72.4	70.6	67.9	58.0	48.2	36.8	(29)
(30)			MADB	35.7	35.9	42.2	56.0	71.7	77.5	79.3	77.1	73.2	62.0	49.9	37.6	(30)
(31)		5%	WB	31.6	32.0	36.3	45.7	59.3	67.4	70.2	68.9	65.2	54.7	43.5	33.9	(31)
(32)			MADB	33.0	34.0	39.5	52.8	66.8	74.8	76.6	74.9	70.1	58.1	45.3	34.8	(32)
(33)		10%	WB	27.2	28.9	34.0	42.4	56.3	64.8	68.1	67.0	62.5	51.2	40.4	31.9	(33)
(34)			MADB	28.9	31.4	36.9	48.3	65.1	72.1	74.6	73.2	67.1	54.1	42.7	33.1	(34)
(35)	Mean Daily Temperature Range			MDBR	20.8	23.0	20.9	19.0	22.8	22.0	20.8	20.9	17.3	14.2	16.8	(35)
(36)		5% DB	MCDBR	18.1	18.0	21.9	28.8	31.7	28.5	24.8	24.8	25.7	25.1	19.3	12.8	(36)
(37)			MCWBR	17.6	16.1	16.3	18.7	17.7	15.6	13.1	13.4	15.9	17.8	16.5	12.4	(37)
(38)			MCDBR	17.6	16.6	18.3	24.6	26.4	23.9	21.0	20.6	22.1	21.6	16.4	12.6	(38)
(39)		5% WB	MCWBR	17.6	15.1	14.6	17.5	16.2	14.6	12.5	12.5	15.2	17.0	15.4	12.7	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.263	0.280	0.320	0.346	0.362	0.396	0.406	0.389	0.365	0.335	0.304	0.285	(40)
(41)		taud		2.306	2.248	2.218	2.356	2.409	2.308	2.297	2.347	2.409	2.487	2.477	2.297	(41)
(42)		Ebn at Noon		264	282	285	288	286	276	271	271	268	261	248	238	(42)
(43)		Edh at Noon		23	31	38	36	36	40	40	36	31	25	21	22	(43)
(44)	All-Sky Solar Radiation	RadAvg		444	731	1112	1409	1605	1746	1706	1483	1163	680	392	299	(44)
(45)		RadStd		61	90	90	120	167	111	169	94	97	57	46	29	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	+1.13	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page